



EDUCATION

BTech - Robotics & Autonomous Systems	Plaksha University, Mohali	8.80/10	2027
12th - CBSE	DAV Public School, Haldia	94.4%	2022
10th - CBSE	DAV Public School, Haldia	94.6%	2020

EXPERIENCE

Summer Intern Robotics Lab, Plaksha	June 2025 - Aug 2025
- Developing an autonomous robot for crop inspection, utilising sensors and Machine Learning for leaf-level disease and pest detection in commercial and horticultural crops. - Collaborated on optimising navigation, data analysis, and real-time crop health monitoring to advance precision farming.	
Senior Teaching Assistant Young Technology Scholar, Plaksha	Mar 2025 - Jun 2025
- Fabricated a strain sensor from scratch using conductive materials and other chemicals. - Built multiple mini-projects to explore sensor applications, and compared strain vs flex sensors for performance and responsiveness. - Integrated sensor with a robotic arm to demonstrate prosthetic control.	
Research Intern Center for Thinking, Language & Communication, Plaksha	Jun 2024 - Jun 2025
- Developed a research paper on project-based learning, which is published by ASEE under the supervision of Prof. Prince. - Co-authored paper on experiential learning frameworks and project-based approaches to education.	
Summer Intern Dixon IoT Lab, Plaksha	Jun 2024 - Nov 2024
- Working on advancing solutions for smart traffic and agricultural management systems under the mentorship of Prof. Srinivasan. - Worked with IoT hardware and different networking protocols like MQTT, CoAP, AMQP.	

PROJECTS

AI Website Assistant for Plaksha University Prof. Sobti	Jan2026 - Present
- Built a semantic AI chatbot for Plaksha University's website with deployment planned for the live website. - Implemented a 3-level RAG architecture: FAQ JSON retrieval, local hybrid retrieval using Chroma + BM25, and Tavily official-domain fallback for weak-context queries. - Designed the pipeline for dynamic scraping, content cleaning, reranking, and cost-efficient LLM usage with open-source embeddings.	
MediRep Self	Feb 2026 - March 2026
- Built a full-stack AI assistant for doctors with real-time chat, voice input, and structured clinical outputs. - Designed a production-grade RAG system integrating drug, interaction, and reimbursement data. - Engineered an end-to-end RAG pipeline with custom guardrails and vector database architecture for safe, reliable responses.	
Intrusion Detection System Prof. Sobti	Aug 2025 - Dec 2025
- Designed and implemented a multiclass IDS leveraging Deep Learning on tabular network traffic data. - Explored various architectures, including LSTM, CNN, Autoencoders, Knowledge Distillation and Transformer, achieving best performance with a hybrid Encoder-Decoder → X-LSTM → CNN pipeline for spatial-temporal pattern learning. - Reached 98.1% accuracy and F1-score with the final model.	
Crop Health Analysis Prof. Siddharth	Feb 2025 - May 2025
- Developed a Deep Learning model for multi-crop health classification using satellite imagery and weather data. - Applied CNN for spatial-spectral feature extraction and LSTM for time-series analysis, comparing ML and DL models on multi-spectral data performance. - Reached 80% accuracy and F1-score with the final model.	

POSITIONS OF RESPONSIBILITY

Class Representative, RAS Plaksha University
Liaised between students and faculty, addressing concerns and ensuring effective communication.
Environmental Captain DAV Public School
Led green initiatives and environmental campaigns within school.

SKILLS

Programming Languages: C++, C, Python, MATLAB Web Development: HTML, CSS, JS IoT: Raspberry Pi, Networking Protocols, ESP32 Technologies: Machine Learning, Deep Learning, Gen AI, RAG, MySQL, MCP, Reinforcement Learning Framework: PyTorch, LangChain, LangGraph, FastAPI

ACHIEVEMENTS

Runners-up, Unstop Hackathon – Developed a soil health monitoring system using IoT and ML for real-time agricultural insights (received incubation opportunity from IIT Ropar). Co-founder, UnDeny – Built an AI-powered platform to help families claim denied medical insurance; ranked Top 4 out of 25 teams. Certificate of Distinction- Australian National Chemistry Quiz, Royal Australian Chemical Institute (International Event)
